

Outlier Detection: Applications And Techniques

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Abstract

Outliers once upon a time regarded as noisy data in statistics, has turned out to be an important problem which is being researched in diverse fields of research and application domains. Many outlier detection techniques have been developed specific to certain application domains, while some techniques are more generic. Some application domains are being researched in strict confidentiality such as research on crime and terrorist activities. The techniques and results of such techniques are not readily forthcoming. A number of surveys, research and review articles and books cover outlier detection techniques in machine learning and statistical domains individually in great details. In this paper we make an attempt to bring together various outlier detection techniques, in a structured and generic description. With this exercise, we hope to attain a better understanding of the different directions of research on outlier analysis for ourselves as well as for beginners in this research field who could then pick up the links to different areas of applications in details.

Keywords: Outlier Applications, Outliers, Outlier Detection.

1. Introduction

Outlier detection aims to find patterns in data that do not conform to expected behavior. It has extensive use in a wide variety of applications such as military surveillance for enemy activities, intrusion detection in cyber security, fraud detection for credit cards, insurance or health care and fault detection in safety critical systems. Their importance in data is due to the fact that they can translate into actionable information in a wide variety of applications. An anomalous traffic pattern in a computer network could mean that a hacked computer is sending out sensitive data to an unauthorized destination [1]. An abnormal MRI image may indicate presence of malignant tumors [2]. Outliers in credit card transaction data could indicate credit card or identity theft [3] or abnormal readings from a space craft sensor could signify a fault in some component of the space craft [4]. In statistical data study of outliers dates as early as the 19th century [5]. Since then several research communities have developed a variety of outlier detection techniques with many of these

specifically meant for certain applications and others being generic in nature. With this exercise, we hope to get a better understanding of the different directions of research on outlier analysis and think of applying techniques in different areas to our areas of interest of crime detection and counter terrorism, even if they were they were not intended, to begin with.

2. Defining Outliers

Outliers are patterns in data that do not conform to a well defined notion of normal behavior. Figure 1 illustrates outliers in a simple 2-dimensional data set. The data has two normal regions, N_1 and N_2 , since most observations lie in these two regions. Points that are sufficiently far away from the regions, e.g., points o_1 and o_2 , and points in region O_3 , are outliers. x y N_1 N_2 o_1 o_2 O_3

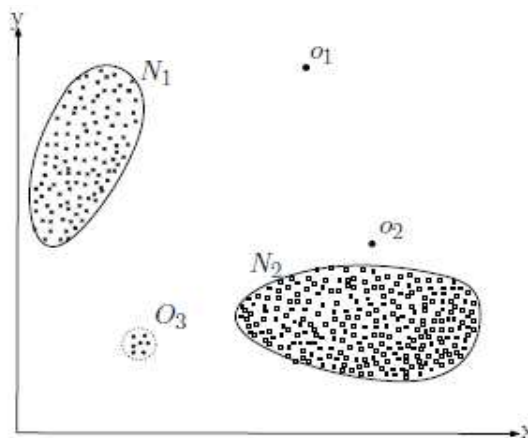


Fig. 1. A simple example of outliers in a 2-dimensional data set.

outliers might be induced in the data for a variety of reasons, such as malicious activity, e.g., credit card fraud, cyber-intrusion, terrorist activity or breakdown of a system,

but the common point of all is that they are interesting to the analyst. The “interestingness” or real life relevance of outliers is a key feature of outlier detection.

Outlier detection is related to, but distinct from noise removal [6] and noise accommodation [7], both of which deal with unwanted noise in the data. Noise can be defined as a phenomenon in data which is not of interest to the analyst, but acts as a hindrance to data analysis. Noise removal is driven by the need to remove the unwanted objects before any data analysis is performed on the data. Noise accommodation refers to immunizing a statistical model estimation against anomalous observations [8].

Another topic related to outlier detection is novelty detection [9,10,11] which aims at detecting previously unobserved (emergent, novel) patterns in the data, e.g., a new topic of discussion in a news group. The distinction between novel patterns and outliers is that the novel patterns are typically incorporated into the normal model after being detected. Another topic related to outlier detection is novelty detection [9,10,11]. The distinction between novel patterns and outliers is that the novel patterns are previously unobserved and get typically incorporated into the normal model after being detected e.g., a new topic of discussion in a news group.

We have discussed above mentioned related problems because their solutions are often used for outlier detection and vice-versa.

3. Difficulties in Outlier Detection

Abstractly speaking outliers are patterns that deviate from expected normal behavior, which in its simplest form could be represented by a region and visualize all normal observations to belong to this normal region and consider the rest as outliers This approach looks simple but is highly challenging due to following reasons.

It is very difficult to define the normal behavior or a normal region. The difficulties are as under.

- Encompassing of every possible normal behavior in the region.
- Imprecise boundary between normal and outlier behavior since at times outlier observation lying close to the boundary could actually be normal, and vice-versa.
- Adaptation of malicious adversaries to make the outlier observations appear like normal when outliers result from malicious actions.

- In many domains normal behavior keeps evolving and may not be current to be a representative in the future.
- Differing notion of outliers in different application domains makes it difficult to apply technique developed in one domain to another. For example, in the medical domain a small deviation from normal body temperature might be an outlier, while similar value deviation in the stock market domain might be considered as normal. Even within same domain say crime detection there could be situations where use of foreign make weapons may be considered normal in crimes committed in metro cities but an outlier for murders of commoners in tribal regions.
- Availability of labeled data for training/ validation of models used by outlier detection techniques.
- Noise in the data which tends to be similar to the actual outliers and hence difficult to distinguish and remove.

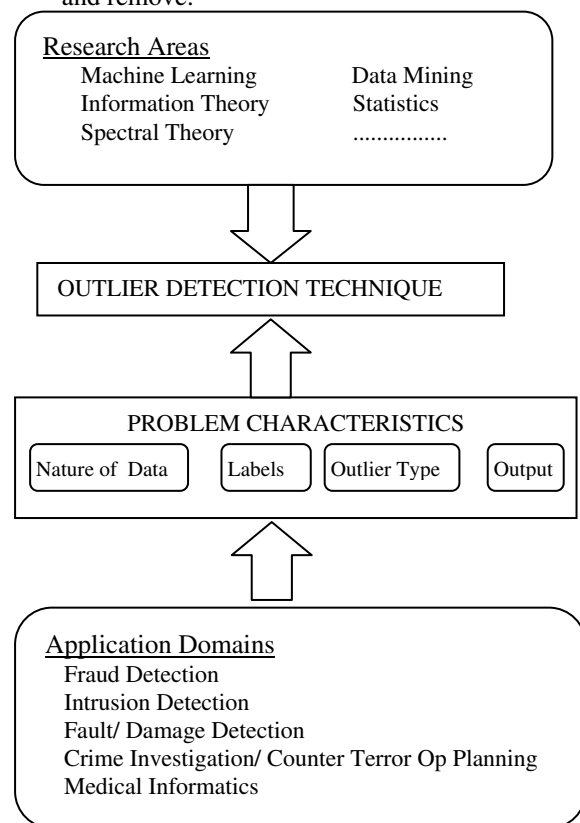


Fig. 2. Key components associated with outlier detection technique.

Due to the above challenges, the outlier detection problem, in its most general form, is not easy to solve. In fact, most of the existing outlier detection techniques solve a specific problem formulation which is induced by various factors

such as nature of the data, availability of labeled data, type of outliers to be detected, etc. Often, these factors are determined by the application domain in which the outliers need to be detected.

Researchers adopt concepts from diverse disciplines such as statistics, machine learning, data mining, information theory, spectral theory, and apply them to specific problem formulations. Figure 2 shows the above mentioned key components associated with any outlier detection technique.

4. Previous Work

A number of surveys, review articles and books especially Hodge and Austin [12], cover outlier detection techniques in machine learning and statistical domains. Numeric and symbolic data approaches [13], neural networks and statistical approaches [9, 10, 11] have been presented by various researchers. Cyber-intrusion detection surveys [14, 15] and research and review books on outlier detection techniques [16,17,18] are excellent sources of literature on the subject.

5. Our Contribution

The above literature on outlier detection set focus on individual applications or on a particular research area. We have attempted to structure and present a broad overview of the detailed research on outlier detection techniques in multifarious research areas and applications also trying to highlight the richness and complexity associated with each application domain. We distinguish simple outliers from complex outliers and define two types of complex outliers, viz., contextual and collective outliers.

6. Aspects Determining the Formulation of Problem

As mentioned earlier, a specific formulation of the problem is determined by several different factors some of which are discussed below. These are also depicted in Figure 2 above. Broadly speaking they are:-

- Nature of Input Data
- Type of Outlier – Point, Contextual, Collective
- Data Labels
- Output of Outlier Detection.

7. Nature of Input Data

This is a key aspect of any outlier detection technique. Input is generally a collection of data instances (also referred as object, record, point, vector, pattern, event, case, sample, observation, entity) [20]. Each data instance can be described using a set of attributes (also referred to as variable, characteristic, feature, field, dimension). The attributes can be of different types such as binary, categorical or continuous. Each data instance might consist of only one attribute (univariate) or multiple attributes (multivariate). In the case of multivariate data instances, all attributes might be of same type or might be a mixture of different data types. The nature of attributes determines the applicability of outlier detection techniques. For example, for statistical techniques different statistical models have to be used for continuous and categorical data. Similarly, for nearest neighbor based techniques, the nature of attributes would determine the distance measure to be used. Often, instead of the actual data, the pair-wise distance between instances might be provided in the form of a distance (or similarity) matrix. In such cases, techniques that require original data instances are not applicable, e.g., many statistical and classification based techniques. In case these statistical methods are applied to OLAP cubes for data mining then the distance between dimensional data can be found out by applying score functions.

Input data can also be categorized based on the relationship present among data instances [20]. Most of the existing outlier detection techniques deal with record data (or point data), in which no relationship is assumed among the data instances. In case these statistical methods are applied to OLAP cubes for data mining then the distance between dimensional data can be found out by applying some sort of score functions and then determining the outliers.

In general, data instances can be related to each other. Some examples are sequence data, spatial data, and graph data. In sequence data, the data instances are linearly ordered, e.g., time-series data, genome sequences, protein sequences. In spatial data, each data instance is related to its neighboring instances, e.g., vehicular traffic data, ecological data. When the spatial data has a temporal (sequential) component it is referred to as spatio-temporal data, e.g., climate data. In graph data, data instances are represented as vertices in a graph and are connected to other vertices with edges. Later we will discuss situations where such relationship among data instances becomes relevant for outlier detection.

8. Types of Outliers

An important aspect of an outlier detection technique is the nature of the desired outlier. Outliers can be classified into following three categories:

- Point Outliers
- Contextual Outliers
- Collective Outliers.

9. Point Outliers

If an individual data instance can be considered as anomalous with respect to the rest of data, then the instance is termed as a point outlier. This is the simplest type of outlier and is the focus of majority of research on outlier detection. For example, in Figure 1, points o1 and o2 as well as points in region O3 lie outside the boundary of the normal regions, and hence are point outliers since they are different from normal data points. As a real life example, if we consider credit card fraud detection with data set corresponding to an individual's credit card transactions assuming data definition by only one feature: amount spent. A transaction for which the amount spent is very high compared to the normal range of expenditure for that person will be a point outlier.

10. Contextual Outliers

If a data instance is anomalous in a specific con-text (but not otherwise), then it is termed as a contextual outlier (also referred to as conditional outlier [21]). The notion of a context is induced by the structure in the data set and has to be specified as a part of the problem formulation. Each data instance is defined using two sets of attributes:

- **Contextual attributes.** The contextual attributes are used to determine the context (or neighborhood) for that instance. For example, in spatial data sets, the longitude and latitude of a location are the contextual attributes. In time-series data, time is a contextual attribute which determines the position of an instance on the entire sequence.
- **Behavioral attributes.** The behavioral attributes define the non-contextual characteristics of an instance. For example, in a spatial data set describing the average rainfall of the entire world, the amount of rainfall at any location is a behavioral attribute.

The anomalous behavior is determined using the values for the behavioral attributes within a specific context. A data instance might be a contextual outlier in a given context, but an identical data instance (in terms of behavioral attributes) could be considered normal in a different context. This property is key in identifying contextual and behavioral attributes for a contextual outlier detection technique.

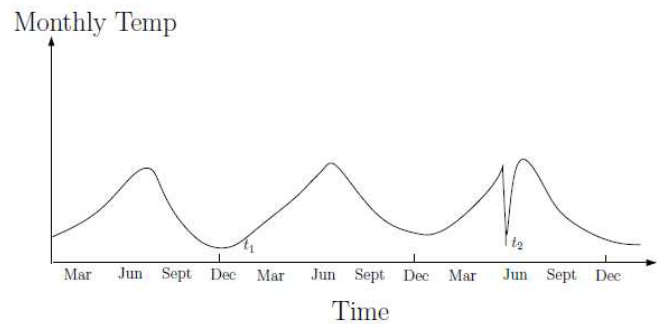


Fig. 3. Contextual outlier t2 in a temperature time series. Temperature at time t1 is same as that at time t2 but occurs in a different context and hence is not considered as an outlier.

Contextual outliers have been most commonly explored in time-series data [22] and spatial data [23]. Figure 3 shows one such example for a temperature time series which shows the monthly temperature of an area over last few years. A temperature of 35F might be normal during the winter (at time t_1) at that place, but the same value during summer (at time t_2) would be an outlier. A six ft tall adult may be a normal person but if viewed in *context of age* a *six feet tall kid* will definitely be an outlier.

A similar example can be found in the credit card fraud detection with contextual as *time* of purchase. Suppose an individual usually has a weekly shopping bill of \$100 except during the Christmas week, when it reaches \$1000. A new purchase of \$1000 in a week in July will be considered a contextual outlier, since it does not conform to the normal behavior of the individual in the context of time (even though the same amount spent during Christmas week will be considered normal).

The choice of applying a contextual outlier detection technique is determined by the meaningfulness of the contextual outliers in the target application domain. Applying a contextual outlier detection technique makes sense if contextual attributes are readily available and therefore defining a context is straightforward. But it becomes difficult to apply such techniques if defining a context is not easy.

11. Collective Outliers

If a collection of related data instances is anomalous with respect to the entire data set, it is termed as a collective outlier. The individual data instances in a collective outlier may not be outliers by themselves, but their occurrence together as a collection is anomalous. Figure 4 illustrates an example which shows a human electrocardiogram output [24]. The highlighted region denotes an outlier because the same low value exists for an abnormally long time (corresponding to an Atrial Premature Contraction). It may be noted that low value by itself is not an outlier but its successive occurrence for long time is an outlier.

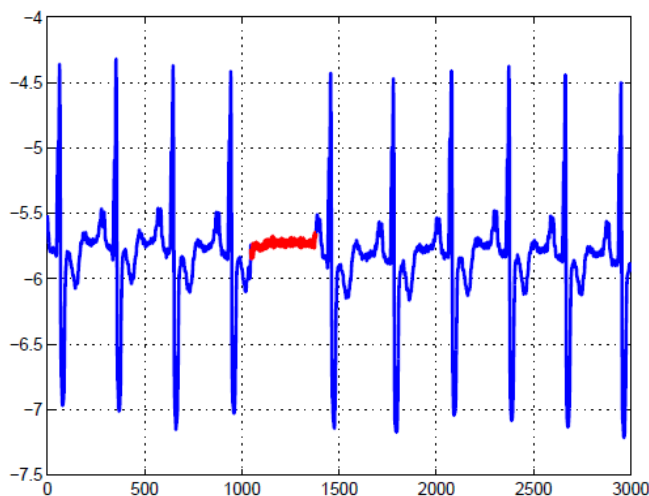


Fig. 4. Collective outlier in an human ECG output corresponding to an Atrial Premature Contraction.

As an another illustrative example, consider a sequence of actions occurring in a computer as shown below:

.....http-web, buffer-overflow, http-web, http-web, smtp-mail, ftp, http-web, ssh, smtp-mail, http-web, ssh, buffer-overflow, ftp, http-web, ftp, smtp-mail, http-web.....

The highlighted sequence of events (buffer-overflow, ssh, ftp) correspond to a typical web based attack by a remote machine followed by copying of data from the host computer to remote destination via ftp. It should be noted that this collection of events is an outlier but the individual events are not outliers when they occur in other locations in the sequence.

Collective outliers have been explored for sequence data [25,26], graph data [27], and spatial data [28]. It should be noted that while point outliers can occur in any data set, collective outliers can occur only in data sets in which data instances are related. In contrast, occurrence of contextual outliers depends on the availability of context attributes in

the data. A point outlier or a collective outlier can also be a contextual outlier if analyzed with respect to a context. Thus a point outlier detection problem or collective outlier detection problem can be transformed to a contextual outlier detection problem by incorporating the context information

12. Data Labels

The labels associated with a data instance denote if that instance is normal or anomalous. It should be noted that obtaining labeled data which is accurate as well as representative of all types of behaviors, is often prohibitively expensive. Labeling is often done manually by a human expert and hence requires substantial effort to obtain the labeled training data set. Typically, getting a labeled set of anomalous data instances which cover all possible type of anomalous behavior is more difficult than getting labels for normal behavior. Moreover, the outlier behavior is often dynamic in nature, e.g., new types of outliers might arise, for which there is no labeled training data. In certain cases, such as air traffic safety, outlier instances would translate to catastrophic events, and hence will be very rare. Based on the extent to which the labels are available, outlier detection techniques can operate in one of the following three modes:

- **Supervised outlier detection:** Techniques trained in supervised mode assume the availability of a training data set which has labeled instances for normal as well as outlier class. Typical approach in such cases is to build a predictive model for normal vs. outlier classes. Any unseen data instance is compared against the model to determine which class it belongs to. There are two major issues that arise in supervised outlier detection. First, the anomalous instances are few, as compared to the normal instances in the training data. Second, obtaining accurate and representative labels, especially for the outlier class is usually challenging. A number of techniques have been proposed [29, 30, 31] that inject artificial outliers in a normal data set to obtain a labeled training data set. Other than these two issues, the supervised outlier detection problem is similar to building predictive models. Hence we will not address this category of techniques in this survey.
- **Semi-Supervised outlier detection:** Techniques that operate in a semi-supervised mode, assume that the training data has labeled instances for only the normal class. Since they do not require labels for the outlier class, they are more widely

applicable than supervised techniques. For example, in space craft fault detection [32], an outlier scenario would signify an accident, which is not easy to model. The typical approach used in such techniques is to build a model for the class corresponding to normal behavior, and use the model to identify outliers in the test data. A limited set of outlier detection techniques exist that assume availability of only the outlier instances for training [25, 33, 34]. Such techniques are not commonly used, primarily because it is difficult to obtain a training data set which covers every possible anomalous behavior that can occur in the data.

- **Unsupervised outlier detection:** Techniques that operate in unsupervised mode do not require training data, and thus are most widely applicable. The techniques in this category make the implicit assumption that normal instances are far more frequent than outliers in the test data. If this assumption is not true then such techniques suffer from high false alarm rate.

Many semi-supervised techniques can be adapted to operate in an unsupervised mode by using a sample of the unlabeled data set as training data. Such adaptation assumes that the test data contains very few outliers and the model learnt during training is robust to these few outliers.

13. Output of Outlier Detection

An important aspect for any outlier detection technique is the manner in which the outliers are reported. Typically, the outputs produced by outlier detection techniques are one of the following two types:

- **Scores:** Scoring techniques assign an outlier score to each instance in the test data depending on the degree to which that instance is considered an outlier. Thus the output of such techniques is a ranked list of outliers. An analyst may choose to either analyze top few outliers or use a cut-off threshold to select the outliers.
- **Labels:** Techniques in this category assign a label (normal or anomalous) to each test instance.

Scoring based outlier detection techniques allow the analyst to use a domain-specific threshold to select the most relevant outliers. Techniques that provide binary labels to the test instances do not directly allow the analysts to make such a choice, though this can be

controlled indirectly through parameter choices within each technique.

14. Applications of Outlier Detection

We shall highlight several applications of outlier detection. For each application we shall discuss following aspects:

- The notion of outlier.
- Nature of the data.
- Challenges associated with detecting outliers.
- Existing outlier detection techniques.

15. Intrusion Detection

Intrusion detection refers to detection of malicious activity (break-ins, penetrations, and other forms of computer abuse) in a computer related system [35] interesting from a computer security perspective. Being different from normal system behavior, intrusion detection is a perfect candidate for applying outlier detection techniques. The key challenges for outlier detection are :-

- **Huge Data Volume:** This calls for computationally efficient techniques.
- **Streaming Data:** This requires on-line analysis.
- **False alarm rate:** Smallest percentage of false alarms among millions of data objects can make be overwhelming for an analyst.
- **Labeled data not usually available for Intrusions:** This gives preference to semi-supervised and unsupervised outlier detection techniques.

Intrusion detection systems have been classified into host based and network based intrusion detection systems [36].

The major differences being tabled as under:-

Table 1: Differences in Nature of Host Based and Network Intrusion Systems

<i>Aspect</i>	<i>Host Based</i>	<i>Network Based</i>
Outliers In	OS Calls	Network Data.
Translates to	Malicious Code Unusual Behaviour Policy Violations	Denial of Network Services
Nature of Data analysis	Sequential	Point, Sequential, Collective
Granularity/ Profiling	User / Program	Packet Level/ NetFlows

The examples of outlier detection techniques for Intrusion Detection are tabled below:-

Table 2: Some outlier detection techniques used in Host Based and Network Intrusion Systems

<i>Technique Used</i>	<i>References</i>
Host Based Intrusion Detection Systems	
Statistical Profiling Using Histograms	[37- 45]
Mixture of Models	[46]
Neural Networks	[47]
Support Vector Machines	[2]
Rule Based Systems	[48 - 50]
Network Based Intrusion Detection Systems	
Statistical Profiling using Histograms	[51 - 54]
Parametric Statistical Modeling	[55]
Non-parametric Statistical Modeling	[56]
Bayesian Networks	[57 - 60]
Support Vector Machines	[46]
Rule Based Systems	[61]
Neural Networks	[46, 62 - 68]

16. Fraud Detection

Fraud refers to criminal activities occurring in commercial organizations such as banks, credit card companies,

insurance agencies, cell phone companies, stock market, etc. Malicious users could be actual customers of the organization or resorting to identity theft (posing as customers). The detection activity aims at detection of unauthorized consumption of resources provided by the organization to prevent economic losses.

A general approach to outlier detection here would involve maintaining a usage profile for each customer and monitor the profiles to detect any deviations termed as activity monitoring [73]. Some specific applications of fraud detection are discussed below.

Credit Card Fraud Detection: Outlier detection techniques are applied to detect :-

- **Fraudulent Applications for Credit Card:** This is similar to detecting insurance fraud [69]
- **Fraudulent Usage of Credit Card:** Associated with credit card thefts.

The data records are defined over several dimensions such as the user ID, spent amount, time between consecutive card usage, etc. The frauds are typically reflected in transactional records (point outliers) and correspond to high payments, high rate of purchase, purchase of items never purchased by the user before, etc. Availability of labeled records is no problem since credit companies have complete data available. Moreover, the data falls into distinct profiles based on the credit card user. Hence profiling and clustering based techniques are typically used in this domain.

Online detection of fraud as soon as fraudulent transaction occurs is a challenge in detecting unauthorized credit card usage. This problem is addressed in two different ways.

Table 3: Approaches in detecting fraudulent transactions.

<i>Approach</i>	<i>By-Owner</i>	<i>By-Operation</i>
Context	User	Geographic Location
Cost	Expensive; querying a central data repository with every transaction.	

Some outlier detection techniques used in fraud detection are listed in Table IV.

Table IV: Some outlier detection techniques used in fraud detection

<i>Technique Used</i>	<i>References</i>
Neural Networks	[3, 69 - 71]
Rule-based Systems	[70]
Clustering	[72]

17. Mobile Phone Fraud Detection.

In this activity monitoring problem the calling behavior of each account is scanned to issue an alarm when an account appears to have been misused.

Calling activity is usually represented with call records. Each call record is a vector of continuous (e.g., Call-Duration) and discrete (e.g., Calling-City) features. However, there is no inherent primitive representation in this domain. Calls are aggregated by time, for example into call-hours or call-days or user or area depending on the granularity desired. The outliers correspond to high volume of calls or calls made to unlikely destinations.

Some techniques applied to cell phone fraud detection are listed in Table V.

Table V: Examples of different outlier detection techniques used for cell phone fraud detection.

<i>Technique Used</i>	<i>References</i>
Statistical Profiling using Histograms	[73, 74]
Parametric Statistical Modelling	[75, 76]
Neural Networks	[77, 78]
Rule based Systems	[78,79]

18. Insurance Claim Fraud Detection

An important problem in the property-casualty insurance industry is claims fraud, e.g. automobile insurance fraud. Individuals and conspiratorial rings of claimants and providers manipulate the claim processing system for unauthorized and illegal claims.

The data in this domain for fraud detection comes from the documents submitted by the claimants. The

techniques extract different features (both categorical as well as continuous) from these documents. Typically, claim adjusters and investigators assess these claims for frauds. These manually investigated cases are used as labeled instances by supervised and semi-supervised techniques for insurance fraud detection.

Insurance claim fraud detection is quite often handled as a generic activity monitoring problem [73]. Neural network based techniques have also been applied to identify anomalous insurance claims [80, 81].

19. Insider Trading Detection

Insider trading is a phenomenon found in stock markets, where people make illegal profits by acting on (or leaking) inside information before the information is made public.

The inside information can be of different forms [82] generally referring to any information which would affect the stock prices in a particular industry. It could be knowledge about a pending merger/acquisition, a terrorist attack affecting a particular industry, a pending legislation affecting a particular industry.

Fraud has to be detected in an online manner and as early as possible, to prevent people/organizations from making illegal profits. The available data comes from heterogeneous sources such as option trading data, stock trading data, news. The data has temporal associations since the data is collected continuously. The temporal and streaming nature has also been exploited in certain techniques [75].

Some outlier detection techniques used in this domain are listed in Table VI.

Table VI: Examples of different outlier detection techniques used for insider trading detection.

<i>Technique Used</i>	<i>References</i>
Statistical profiling using Histograms	[75, 82]
Information Theoretic	[83]

20. Medical and Public Health Outlier Detection

The data typically consists of patient records which may have several different types of features such as patient age, blood group, weight. The data might also

have temporal as well as spatial aspect to it. The data can have outliers due to several reasons such as abnormal patient condition or instrumentation errors or recording errors. Most of the current outlier detection techniques in this domain aim at detecting anomalous records (point outliers). Typically the labeled data belongs to the healthy patients, hence most of the techniques adopt semi-supervised approach. Another form of data handled by outlier detection techniques in this domain is time series data, such as Electrocardiograms (ECG) and Electroencephalograms (EEG). Collective outlier detection techniques have been applied to detect outliers in such data [91]. Several techniques have also focussed on detecting disease outbreaks in a specific area [90]. Thus the outlier detection is a very critical problem in this domain and requires high degree of accuracy.

The most challenging aspect of the outlier detection problem in this domain is that the cost of classifying an outlier as normal can be very high.

Some outlier detection techniques used in this domain are listed in Table VII.

Table VII: Examples of different outlier detection techniques used in medical and public health domain.

<i>Technique Used</i>	<i>References</i>
Parametric Statistical Modelling	[84 - 88]
Neural Networks	[89]
Bayesian Networks	[90]
Rule-based Systems	[75]
Nearest Neighbor based techniques	[91]

21. Industrial Damage Detection

Industrial units suffer damage due to continuous usage and the normal wear and tear. Such damages need to be detected early to prevent further escalation and losses. The data in this domain is usually sensor data recorded using different sensors and collected for analysis.

Outlier detection in this domain is classified into two fields as tabulated below.

Table VIII: Characteristics of Fault Detection in Mechanical Units and Structural Damage Domain.

<i>Aspect</i>	<i>System Health Management</i>	<i>By-Operation</i>
Defects Dealt pertaining to	Mech components such as motors, engines, turbines, oil flow in pipelines etc.	Structures,
Cause of Defects	Wear and Tear or other unforeseen circumstances.	Cracks in beams, strains in airframes . Unforeseen data.
Data Aspect	Temporal	Temporal
Analysis	Time Series	Time series with special correlations
Types of Outliers	Contextual or Collective outliers	Novelty detection or change point detections
Normal data	Readily Available	Is learnt and typically static over time.
Supervision	Semi-supervised	Semi-supervised
Literature	[94, 95]	[108, 111, 112, 115]
Techniques	Table IX.	Table X.

Table IX: Examples of outlier detection techniques used for fault detection in mechanical units.

<i>Technique Used</i>	<i>References</i>
Parametric Statistical Modelling	[92, 93, 94, 95]
Non-Parametric Statistical Modelling	[96]
Neural Networks	[97, 89, 98-105]
Spectral	[4, 106]
Rule Based Systems	[107]

Table X: Examples of outlier detection techniques used for structural damage detection.

<i>Technique Used</i>	<i>References</i>
Statistical profiling using Histograms	[108, 109, 110]
Parametric Statistical Modelling	[111]
Mixture of Models	[112, 113, 114]
Neural Networks	[115 to 122]

22. Image Processing

Outlier detection here aims to detect changes in an image over time (motion detection) or in regions which appear abnormal on the static image. This domain includes satellite imagery, digit recognition, spectroscopy, mammographic image, and video surveillance. The outliers are caused by motion or insertion of foreign object or instrumentation errors. The data has spatial as well as temporal characteristics. Each data point has a few continuous attributes such as color, lightness, texture, etc. The interesting outliers are either anomalous points or regions in the images (point and contextual outliers).

One of the key challenges in this domain is the large size of the input. The challenge is greater when dealing with video data and, online detection techniques are required.

Some references on various applications are tabulated below:-

Table XI: Examples of outlier detection techniques used in image processing domain.

<i>Application Domain</i>	<i>References</i>
Satellite Imagery	[123, 124, 125, 126, 127]
Digit Recognition	[128]
Mammographic Image Analysis	[129, 130]
Spectroscopy	[131, 132, 133, 134]
Video Surveillance	[135, 136, 137].

Some outlier detection techniques used in this domain are listed in Table XII.

Table XII: Examples of outlier detection techniques used in image processing domain.

<i>Technique Used</i>	<i>References</i>
Mixture of Models	[124, 129, 130]
Regression	[126, 131]
Bayesian Networks	[135]
Support Vector Machines	[132, 138]
Neural Networks	[123, 125, 128, 133, 136]
Clustering	[134]
Nearest Neighbour Techniques	[124, 137]

23. Outlier Detection in Text Data

Outlier detection techniques in this domain primarily detect novel topics or events or news stories in a collection of documents or news articles. The outliers are caused due to a new interesting event or an anomalous topic. The data in this domain is typically high dimensional and very sparse. The data also has a temporal aspect since the documents are collected over time.

A challenge for outlier detection techniques in this domain is to handle the large variations in documents belonging to one category or topic. Some outlier detection techniques used in this domain are listed in Table XIII.

Table XIII: Examples of techniques used for outlier topic detection in text data.

<i>Technique Used</i>	<i>References</i>
Statistical Profiling using Histograms	[73]
Mixture of Models	[139]
Neural Networks	[140]
Support Vector Machines	[141]
Clustering Based	[142, 143, 144]

24. Sensor Networks

Sensor networks have lately become an important topic of research from data analysis perspective, since the data collected from various wireless sensors has several unique characteristics. Outliers in such data collected can either imply one or more faulty sensors (sensor fault detection applications), or the sensors are detecting events (intrusion detection applications) that are interesting for analysts.

A single sensor network might comprise a mix of sensors that collecting different types of data, such as binary, discrete, continuous, audio, video, etc. The data is generated in a streaming mode and the collected data often contains noise and missing values due to limitations imposed by deployment environment and communication channel.

This poses a set of unique challenges. The streaming data calls for outlier detection techniques to operate in an online approach. The severe resource

constraints call for light-weight detection techniques. The data collected in a distributed fashion calls for a distributed data mining approach to analyze the data [145]. Lastly the presence of noise in sensor data makes outlier detection more challenging, since it has to now distinguish between interesting outliers and the unwanted values (noise/missing values).

Table XIV lists some outlier detection techniques used in this domain.

Table XIV: Some outlier detection techniques used for outlier detection in sensor networks.

<i>Technique Used</i>	<i>References</i>
Bayesian Networks	[146]
Rule-based Systems	[147]
Parametric Statistical Modelling	[148, 149]
Nearest Neighbor Based Techniques	[150, 151, 152]
Spectral Techniques	[145]

25. Other Domains

Some other domains where outlier detection has also been applied are as tabulated below.

Table XV: Examples of outlier detection techniques used in other application domains.

<i>Technique Used</i>	<i>References</i>
Speech Recognition	[153, 154]
Novelty Detection in Robot Behavior	[155, 156, 157, 158, 159]
Traffic Monitoring	[160]
Click Through Protection	[161]
Detecting Faults in Web Applications	[162, 163]
Detecting Outliers in Biological Data	[55, 164, 165, 166, 167, 176]
Detecting Outliers in Census Data	[168]
Detecting Associations among Criminal Activities	[169]

Detecting Outliers in Customer Relationship Management (CRM) Data	[170]
Detecting Outliers in Astronomical Data	[171, 172, 173]
Detecting Ecosystem Disturbances	[23,174, 175]

26. Conclusion

In this paper we have brought together various outlier detection techniques, in a structured and generic description. With this exercise, we have attained a better understanding of the different directions of research on outlier analysis for ourselves as well as for beginners in this research field who can pick up the links to different areas of applications in details.

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